

SSR: Scaling Surefooted and Symmetric Humanoid Traversal to the Open World

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Figure 1: SSR scales surefooted and symmetric humanoid traversal to diverse open-world human environments. Driven by egocentric vision, the robot reliably traverses grassy slopes, stairs of varying sizes and structures, jumps onto high platforms, and crosses wide gaps. Throughout long-horizon traversal, it maintains safe foot placement together with coordinated and natural whole-body motion.

Abstract: Extending humanoid traversal to open-world human environments is essential for real-world deployment, yet remains challenging. The robot must use vision to ensure safe and reliable foot placement on heterogeneous terrain under highly dynamic motion, while producing coordinated, natural whole-body behaviors. We propose **SSR**, an efficient end-to-end framework for egocentric vision-based humanoid traversal that jointly learns these capabilities. SSR introduces a predictive foothold safety reward that provides dense swing-phase guidance, encouraging earlier commitment to stable footholds to reduce edge slipping. It further employs a compact equivariant latent-space symmetry augmentation to efficiently induce bilateral coordination under high-dimensional visual observations, together with terrain-specific multi-discriminator motion priors that preserve human-like behavior across scenes. Extensive experiments show that SSR transfers safe, stable, and high-quality locomotion to diverse real-world terrains, including stairs with varied structures, and to extreme challenges such as wide gaps and high platforms, while enabling reliable long-horizon traversal in open outdoor environments. Project website: <https://ssr-humanoid.github.io/>.

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1 Introduction

Humanoid robots are naturally suited to complex real-world environments because their morphology and movement resemble those of humans. Realizing this potential requires agile, stable, and natural traversal over heterogeneous terrains, from everyday stairs to wide gaps and high platforms [1]. This demands that the robot perceive changing surroundings, place each step within safe support regions, and preserve human-like whole-body behavior throughout prolonged locomotion [2].

To anticipate terrain changes, recent learning-based humanoid locomotion leverages onboard perception. Among these approaches, policies conditioned on egocentric depth images are especially promising in dynamic real-world scenes as they avoid explicit, noise-sensitive mapping [3, 4, 5, 6]. Yet, turning vision into robust locomotion over open-world terrains remains an open challenge.

The difficulty goes beyond terrain recognition. A perceptive policy must map visual cues into reliable foothold decisions, as stepping near terrain edges sharply reduces support area for flat-footed humanoids and raises the risk of slips or failure. Real-world deployment also demands bilateral coordination and human-like motion. Although prior work has explored symmetry priors [7, 8, 9] and motion imitation [10, 11, 12], integrating them efficiently into egocentric-vision policies remains difficult. High-dimensional visual inputs make symmetry learning costly, while terrain-dependent dynamics reduce the stability of human-like style modeling. Moreover, different terrains often require distinct movement strategies, hindering unified learning in a single policy [13]. Overall, a efficient framework that jointly enables safe footholds, symmetric and natural whole-body behavior, and robust traversal across diverse terrains under egocentric vision is still lacking.

In this paper, we propose **SSR**, a single-stage end-to-end reinforcement learning (RL) framework for vision-based humanoid traversal in challenging real-world environments. It learns directly from depth and proprioception, and uses a Mixture-of-Experts (MoE) architecture to cross-scene policy expressiveness [13, 14]. Within this unified framework, the policy jointly learns reliable foot placement and high-quality whole-body motion, combining gait symmetry with human-like behavior.

Safe, accurate foot placement on complex terrain requires informative reward design. Prior methods often evaluate foothold safety only at or after contact, leaving swing-phase actions with sparse, delayed feedback [3, 4, 15]. We therefore introduce a predictive dense foothold safety reward that predicts the next foothold during swing and assesses its surrounding geometry. This turns a contact-time safety objective into continuous swing-phase guidance, encouraging earlier correction and more proactive targeting of flat, well-supported landing regions to reduce edge contacts, slips, and falls.

We next address whole-body motion quality for perceptive traversal. Existing methods impose symmetry via data augmentation, symmetry regularization, or strict equivariant architecture [8, 9, 16]. For egocentric-vision policies, the first two require extra inference on mirrored inputs and, with memory, additional hidden-state rollouts, whereas the third may yield overly rigid motion due to neutral-state issues [17]. We therefore develop a latent-space symmetry augmentation with an equivariant encoder, moving mirror operations from image input to a compact latent to reduce overhead while preserving flexibility. We further use a terrain-specific multi-discriminator adversarial motion prior [18] to capture terrain-dependent style and maintain human-like motion across scenes.

We evaluate SSR in diverse environments. As shown in Fig. 1, the robot robustly traverses slopes, rough ground, and stairs of varied geometries. It can jump onto platforms up to 45 cm high, about $1.6\times$ its shank length, and cross gaps as wide as 90 cm. More importantly, these capabilities transfer to complex outdoor scenes and support reliable long-horizon locomotion with precise foot placement, strong coordination, good controllability, and natural motion. Our main contributions are:

1. We present a novel single-stage training framework that jointly learns safety, symmetry, and human-like motion for vision-based humanoid traversal in diverse environments.
2. We introduce a predictive foothold safety reward to provides dense swing-phase feedback, enabling earlier correction of unsafe actions and more robust foot placement.
3. We develop a latent-space augmentation with an equivariant encoder for efficient symmetry learning, and combine it with terrain-specific style priors to jointly improve motion quality.
4. We validate the learned policy through extensive indoor and outdoor experiments, demonstrating agile and reliable humanoid locomotion over heterogeneous and challenging open-world settings.

2 Related Works

Learning Perceptive Humanoid Locomotion Data-driven RL has advanced humanoid locomotion under complex contact dynamics and environmental variation [1, 19, 20], and exteroceptive sensing let controllers adapt to terrain before touchdown. Earlier methods construct height maps from LiDAR and odometry [21, 22, 23, 24], but depend on accurate localization and is sensitive to latency, drift, limited update rates, and occlusion. Recent approaches infer terrain directly from egocentric depth images [3, 4, 5, 25, 26], reducing reliance on mapping. However, many vision-based humanoid systems still target structured settings or require multi-stage pipelines, expert distillation, or dedicated depth modeling and augmentation [5, 20, 27, 28, 29]. In contrast, we learn a unified traversal policy from egocentric vision in a single-stage framework for diverse unstructured terrains.

Safe Foot Placement Classical footstep control decomposes locomotion into perception, foothold planning, and tracking [30, 31, 32]. While effective in specific settings, they rely on hand-crafted design and conservative assumptions. RL-based methods learn foot placement end to end by tracking planner-generated footholds [33, 34, 35, 36] or penalizing unsafe landing regions [3, 4, 15, 37]. Yet these signals are often sparse, appearing only at or after contact, weakening swing-phase guidance and training efficiency [19, 38]. Liu et al. [28] densify learning by swing-trajectory tracking, but still depend on rule-based foothold generation and focus on stairs. Zhu et al. [39] reward derived foothold actions to improve landing quality. We provide dense safety feedback by predicting next foothold during swing, encouraging flatter and safer landings without planners or edge detection.

Symmetry and Human-Like Motion Learning

High-quality locomotion requires coordinated and natural motion. Prior work exploits morphological symmetry via data augmentation or regularization [8, 9, 16, 40], but under visual inputs with short-term memory, these methods require mirrored forward passes on high-dimensional inputs or mirrored hidden-state rollout, increasing computational and memory cost. Strictly equivariant architectures reduce this overhead [7], but may limit symmetry breaking near neutral states for gait exploration [49]. Fig. 2 highlights these trade-offs and motivates our latent-space augmentation for efficient yet flexible symmetry learning.

Complementary to gait coordination, human-like motion learning commonly uses reference tracking [20, 26, 41, 42, 43] or adversarial motion priors [10, 11, 12, 44]. Since a single discriminator may suffer from mode collapse or style mixing [45], we follow Vollenweider et al. [18] and adopt terrain-specific multi-discriminators to maintain natural motion across terrains.

3 Method

Our goal is safe and high-quality humanoid traversal over diverse unstructured terrains directly from proprioception and raw depth in a unified, efficient framework. SSR achieve this with a predictive dense foothold safety reward for earlier swing-phase correction (Sec. 3.2), an equivariant encoder for efficient bilateral gait learning in a compact latent space (Sec. 3.3), and cross-terrain motion priors for stable, human-like whole-body behavior (Sec. 3.4). Fig. 3 gives an overview of SSR.

3.1 Single-Stage Learning of Generalizable Traversal Skills

We formulate humanoid locomotion as a partially observable Markov decision process (POMDP) and optimize the policy with PPO [46] under an asymmetric actor-critic architecture [47].

Observations and Action At timestep t , the policy receives proprioceptive and visual inputs. The proprioception $\mathbf{o}_t^p \in \mathbb{R}^{72}$ includes base angular velocity $\boldsymbol{\omega}_t$, projected gravity $\mathbf{g}_t$, velocity commands $\mathbf{u}_t$, joint positions $\boldsymbol{\theta}_t$, joint velocities $\dot{\boldsymbol{\theta}}_t$, and previous action $\mathbf{a}_{t-1}$. We stack a short history $\mathbf{o}_{t-h+1:t}^p$ to reduce partial observability. Together with the depth image $\mathbf{I}_t \in \mathbb{R}^{36 \times 36}$, this forms the policy observation $\mathbf{o}_t = (\mathbf{o}_{t-h+1:t}^p, \mathbf{I}_t)$, which maps to full-body joint position targets $\mathbf{a}_t \in \mathbb{R}^{21}$. The critic additionally uses privileged information, including ground-truth base velocity $\mathbf{v}_t$, foot velocities $\mathbf{v}_t^f$, contact states $\mathbf{c}_t$, limb positions $\mathbf{p}_t^h$ and $\mathbf{p}_t^f$, and height maps around the feet and body, $\mathbf{H}_t^f$ and $\mathbf{H}_t^b$.

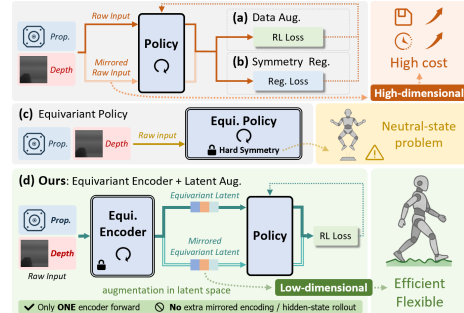


Figure 2: (a-c) Existing symmetry learning methods under visual recurrent policies. (d) Our equivariant latent-space augmentation balances efficiency, flexibility, and gait coordination.

Online Prediction of the Next Foothold During training, we learn a next-foothold predictor that maps the privileged state $\mathbf{s}_t$ and action $\mathbf{a}_t$ to $\hat{\mathbf{F}}_t = \{\hat{F}_{i,t}\}_{i=1}^2$. For foot i , $\hat{F}_{i,t} = (\hat{\boldsymbol{\mu}}_{i,t}, \hat{\sigma}_{i,t})$ parameterizes a Gaussian next-foothold distribution in the current base frame: $q_{i,t}(\mathbf{p}) = \mathcal{N}(\mathbf{p}; \hat{\boldsymbol{\mu}}_{i,t}, (\hat{\sigma}_{i,t})^2 \mathbf{I})$, where $\hat{\boldsymbol{\mu}}_{i,t} \in \mathbb{R}^2$ is the predicted foothold location and $\hat{\sigma}_{i,t} \in \mathbb{R}$ captures uncertainty. This probabilistic form yields a smoother, more stable safety estimate over the predicted landing region. The supervision target $\mathbf{p}_{i,t}^*$ is the first future contact location of foot i , expressed in the current base frame. We train the predictor with the Gaussian negative log-likelihood:

$$\mathcal{L}_{\text{pred}} = - \sum_{i=1}^2 \log q_{i,t}(\mathbf{p}_{i,t}^*) \propto \sum_{i=1}^2 \left(\frac{\|\mathbf{p}_{i,t}^* - \hat{\boldsymbol{\mu}}_{i,t}\|_2^2}{2(\hat{\sigma}_{i,t})^2} + 2\log \hat{\sigma}_{i,t} \right). \quad (3)$$

Early in training, unstable foot placement makes the predictor unreliable, so we use terrain level as a proxy for reliability and enable swing-phase predictive feedback only after it reaches a preset level.

3.3 Efficient Symmetry Learning with Equivariant Latent Augmentation

To impose symmetry priors efficiently under egocentric vision and short-term memory, we perform augmentation in a low-dimensional latent space instead of duplicating high-dimensional inputs. We define mirror transformations, then construct a mirror-equivariant encoder to learn corresponding equivariant latent representation, and generate mirrored samples from that latent space.

Mirror Transformations For a bilaterally symmetric humanoid, let $\mathcal{M}_s$, $\mathcal{M}_o$, and $\mathcal{M}_a$ denote mirror transformations on the state, actor-observation, and action spaces. Since the actor input is $\mathbf{o}_t^a = [\mathbf{o}_t^p, \mathbf{l}_t]^\top$, $\mathcal{M}_o$ is determined by the proprioceptive transform $\mathcal{M}_p$ and latent transform $\mathcal{M}_l$, where $\mathcal{M}_p$ follows physical semantics. For the encoder output $\mathbf{l}_t = [\hat{\mathbf{v}}_t, \hat{\mathbf{z}}_t]^\top$, $\mathcal{M}_l$ consists of $\mathcal{M}_v$ for velocity and $\mathcal{M}_c$ for implicit features. We introduce a symmetry-structured representation that organizes variables in $\hat{\mathbf{z}}_t$ without predefined mirror rules into explicit left-right channel groups, so that $\mathcal{M}_c$ is defined as left-right channel swapping. Appendix B.1 gives the detailed reflection rules.

Equivariant Visual-Memory Encoding To make latent-space augmentation consistent with input mirroring, the encoder must satisfy mirror equivariance: mirrored observations should produce the mirrored latent. Following Cesa et al. [52], we design equivariant linear and convolutional layers, from which we build equivariant MLP, CNN, and GRU modules that together form the encoder.

To map mirrored inputs to mirrored features throughout the network, both the encoder input and output must have explicit left-right structure. The image $\mathbf{I}_t$ naturally satisfies this requirement, with robot-centric horizontal reflection $\mathcal{M}_{2D}\mathbf{I}_t$. In contrast, the frame-wise stacked temporal proprioception $\mathbf{o}_{t-h+1:t}^p$ does not explicitly distinguish left from right, so we introduce a fixed operator $\mathcal{T}$ to organize it into symmetry-structured representation: $\bar{\mathbf{o}}_t^p := \mathcal{T}(\mathbf{o}_{t-h+1:t}^p)$. The encoder then satisfies $E_\phi(\mathcal{M}_c \bar{\mathbf{o}}_t^p, \mathcal{M}_{2D}\mathbf{I}_t) = \mathcal{M}_l E_\phi(\bar{\mathbf{o}}_t^p, \mathbf{I}_t)$. This equivariance compresses high-dimensional visual input and temporal proprioception into a low-dimensional latent space where mirror transformations can be applied directly. Construction details and the proof are provided in Appendix B.2.

Latent-Space Symmetry Augmentation Training needs only one encoder forward pass on original observation. We construct mirrored actor inputs and actions in latent space: $\mathbf{o}_t^{a,g} = \mathcal{M}_o \mathbf{o}_t^a$ and $\mathbf{a}_t^g = \mathcal{M}_a \mathbf{a}_t$. The augmented trajectory $\tau^g = (\mathbf{o}_0^{a,g}, \mathbf{a}_0^g, \dots)$ is used with original trajectory for policy updates [8]. Unlike input-level augmentation, this avoids re-encoding images or collecting hidden states for mirrored rollouts, enabling efficient symmetry learning for visual recurrent policies.

3.4 Cross-Terrain Motion Priors with Multi-Discriminators

To preserve human-like motion across terrains, we employ multiple discriminators [18], one for each terrain type i . Each D_i models terrain-specific reference motion style from a five-frame history ψ_t , where each frame $\mathbf{s}_t^{\text{amp}} \in \mathbb{R}^{63}$ contains projected gravity, base linear and angular velocities, joint position and velocities, and limb positions. Following AMP [10], D_i is trained to distinguish reference from policy motions and produce a style reward that encourages terrain-appropriate natural behavior. We collect motion datasets for each terrain type. See details in Appendix A.8.

4 Experiments

4.1 Experimental Setup

We evaluate SSR in extensive simulation and real-world experiments to answer three questions: (1) Can it learn a unified policy for high-performance traversal over diverse challenging terrains? (2)

Do the key designs improve motion quality and learning efficiency? (3) Can it transfer zero-shot to the unseen open-world scenes with long-horizon robustness and generalization?

Training Environment We perform single-stage training in Isaac Gym [53] with 4,096 AgiBot X2 humanoids on one NVIDIA RTX 4090. We implement warp ray casting to synthesize deployment-consistent depth from terrain and self-occlusion [54]. Appendix A provides details.

Onboard Implementation On the real robot, a waist-mounted Intel RealSense D435i captures forward-facing depth at 60 Hz. Images are downsampled and cropped from 640×360 to 36×36 . An onboard Jetson AGX Orin runs network inference and outputs actions at 50 Hz with onnxruntime.

Compared Baselines We compare SSR with two perceptive baselines: (1) **HPL** [3], a multi-stage parkour method with egocentric vision; and (2) **PIM** [21], a single-stage traversal method with height maps and a hybrid internal model. We further consider the following ablations: (3) **NoFoothold** removes the foothold safety reward r^f ; (4) **NoPred** retains only the sparse touchdown term in r^f and removes predictive dense feedback; (5) **NoSym** disables data augmentation and uses a non-equivariant encoder; (6) **NoEqui** applies data augmentation through input mirroring rather than in equivariant latent space, and uses a non-equivariant encoder; (7) **NoStyle** removes the AMP style reward r^s ; and (8) **SglDisc** replaces terrain-specific multi-discriminator with a shared one.

Evaluation Metric and Protocol We use success rate, defined as the percentage of trials that complete a 20 s traversal without termination, as the primary metric of traversability. In simulation, each method is evaluated over 1,000 randomized trials per terrain under a 1.0 m/s forward command.

4.2 Overall Traversal Performance

Fig. 5 shows that SSR outperforms all baselines. It maintains near-100% success across all training difficulties, even on the hardest stairs, where a 22 cm foot has only a 3 cm support margin. It also traverses a 90 cm gap and a 45 cm platform beyond the curriculum range, demonstrating strong geometric adaptability and extrapolation. By contrast, HPL [3] and PIM [21] degrades rapidly with increasing difficulty, largely because they lack dense foothold safety reward for efficiently learning precise foot placement. Ablation results show that all SSR components matter, with largest drops in NoFoothold and NoPred, highlighting the importance of reliable foot placement for stable traversal.

4.3 Safer Foot Placement via Predictive Dense Reward

We evaluate safe foot placement with two metrics: safe foothold rate (SFR), the fraction of footholds with support ratio above 75%, and mean support ratio (MSR), averaged over footholds. The support ratio is one minus the support deficiency in Sec. 3.2. As shown in Fig. 6, SSR achieves the highest SFR on all terrains, indicating consistently safer contacts. Even on terrains requiring precise landing such as stairs down and gap, it concentrates footholds within valid support regions. The checkpoint-wise MSR curves further show that SSR reaches 75% earliest, reflecting better learning efficiency. In contrast, NoFoothold produces more aggressive gaits and the lowest safety. NoPred relies on sparse post-touchdown feedback, which delays swing-phase correction and still yields edge-biased contacts, as the policy adjusts only when the foot nears the ground. These results highlight the value of predictive dense foothold reward: continuous swing-phase feedback enables earlier trajectory correction, improves credit assignment, and supports more effective learning of safe traversal.

4.4 Coordinated Locomotion via Efficient Latent-Space Symmetry Learning

Fig. 7 (a,b) shows that symmetry augmentation improves controllability and bilateral coordination. With eight-direction velocity commands, SSR tracks more accurately and, under bidirectional yaw-rate commands, stays more localized during in-place turning, with nearly mirror-symmetric trajectories. NoSym instead accumulates drift and left-right imbalance, revealing direction-dependent biases

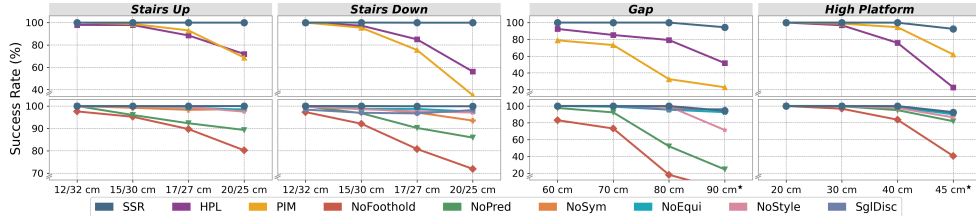


Figure 5: Traversal performance across terrains and difficulty levels in simulation. Top row compares SSR with prior methods, and bottom row reports ablation results. ★ denote difficulties beyond the training curriculum.

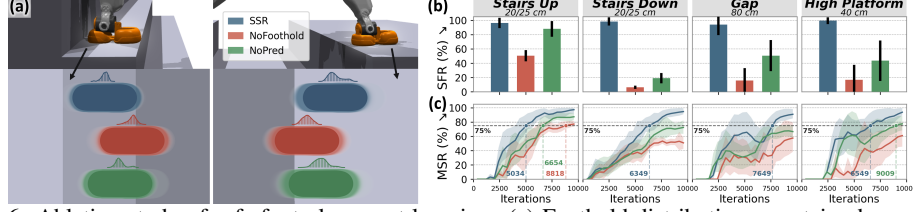


Figure 6: Ablation study of safe foot placement learning. (a) Foothold distributions on stairs down and gap, smoothed with kernel density estimation (KDE). For each terrain, (b) the safe foothold rate (SFR); and (c) the mean support ratio (MSR) versus training iterations, evaluated using checkpoints saved every 100 iterations, with the dashed line marking the first point at which the MSR reaches 75%.

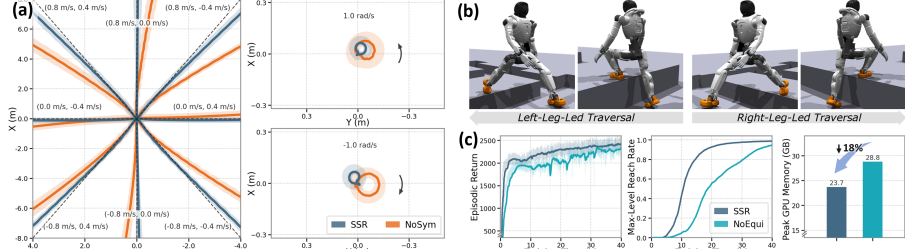


Figure 7: Ablation study of symmetry learning. (a) Trajectories under eight linear-velocity and two in-place yaw-rate commands. (b) SSR shows either-leg-led traversal on gaps and platforms, indicating bilateral balance. (c) Training-time curves of episodic return and maximum-terrain-level reach rate, with peak GPU memory.

and fixed lead-leg dominance. In contrast, SSR supports both left- and right-foot-led traversal, indicating stronger bilateral skill acquisition. Fig. 7 (c) further evaluates training efficiency. Compared with NoEqui, SSR achieves higher returns within the same training time, reaches maximum difficulty earlier, and reduces GPU memory from 28.8 GB to 23.7 GB (18%), making training feasible on one RTX 4090. This gain comes from mirroring a compact latent rather than high-dimensional inputs, avoiding extra encoding passes and hidden-state rollouts. Overall, equivariant latent augmentation scales symmetry-driven high-quality locomotion to visual recurrent policy learning.

4.5 Human-Like Motion via Terrain-Specific AMP

Human-like legged locomotion features low mechanical cost and smooth contacts [55, 56], so we evaluate it using average power and peak foot contact force. Table 1 shows SSR achieves the best overall metrics, with more harmonious whole-body motion, including more natural arm swing and softer impacts, which suggests that terrain-specific multi-discriminator provides motion priors better aligned with terrain dynamics. Fig. 8 illustrates these natural motions across terrains.

Method	Stairs Up		Stairs Down		Gap		High Platform	
	AP(W)	PFCF(N)	AP(W)	PFCF(N)	AP(W)	PFCF(N)	AP(W)	PFCF(N)
SSR	381.3 $\pm$ 5.4	507.7 $\pm$ 12.6	342.4 $\pm$ 8.6	519.2 $\pm$ 19.3	214.4 $\pm$ 5.7	463.4 $\pm$ 15.3	253.0 $\pm$ 6.3	480.4 $\pm$ 14.9
NoStyle	451.6 $\pm$ 10.1	538.5 $\pm$ 13.5	371.4 $\pm$ 43.5	555.5 $\pm$ 21.5	317.0 $\pm$ 6.5	518.1 $\pm$ 16.6	292.3 $\pm$ 6.3	508.7 $\pm$ 16.4
SglDisc	391.8 $\pm$ 5.9	544.7 $\pm$ 9.7	345.3 $\pm$ 24.3	549.4 $\pm$ 23.4	241.1 $\pm$ 7.4	481.8 $\pm$ 14.7	258.2 $\pm$ 5.9	505.8 $\pm$ 16.4

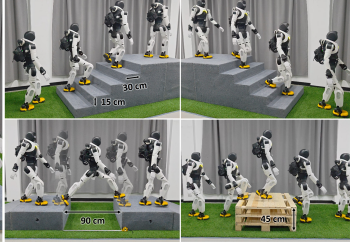
Table 1: Ablation study of human-like motion, measured by average power (AP) and peak foot contact force (PFCF). Results are reported as mean $\pm$ std.

4.6 Real-World Lab-Level Evaluation

We evaluate SSR’s sim-to-real transfer via zero-shot deploying the trained policy on the real robot. Table 2 shows high success rates on all terrains, including the unseen 90 cm gap and 45 cm platform, close to simulation. To our knowledge, these difficulty levels surpass prior single-policy real-world results. Fig. 8 shows consistently precise footholds on flat support regions, coordinated gait, and natural whole-body behavior, which together improve traversal reliability and motion quality.

4.7 Real-World Outdoor Deployment

We validate SSR in extensive outdoor environments, as shown in Fig. 1 and Fig. 9. Over a continuous 1.3 km hike across diverse terrains, the robot maintains safe, coordinated, and natural locomotion, demonstrating long-horizon reliability. Fig. 9 (b-d) further highlights three abilities. **Perturbation Recovery**: the robot remains stable under horizontal and torsional shaking; when pushed from behind during stair descent, it still lands on the flat tread, indicating accurate foothold selection under limited reaction times. **Visual Occlusion Adaptation**: under grass-induced occlusion, it shows no obvious gait degradation and can actively detours poorly observed regions. **Zero-Shot Out-of-**



Terrain	Settings	Success Rate
<i>Standard Settings</i>		
Stairs Up	15 / 30 cm	100.0% (20/20)
Stairs Down	15 / 30 cm	100.0% (20/20)
Gap	80 cm	95.0% (19/20)
Platform	40 cm	100.0% (20/20)
<i>Hard Settings</i>		
Gap-H	90 cm *	85.0% (17/20)
Platform-H	45 cm *	95.0% (19/20)

Figure 8: Key frames of zero-shot real-world lab-level deployment. SSR enables stair traversal, crossing a 90 cm gap, and climbing onto a 45 cm platform with safe foot placement and natural whole-body motion.

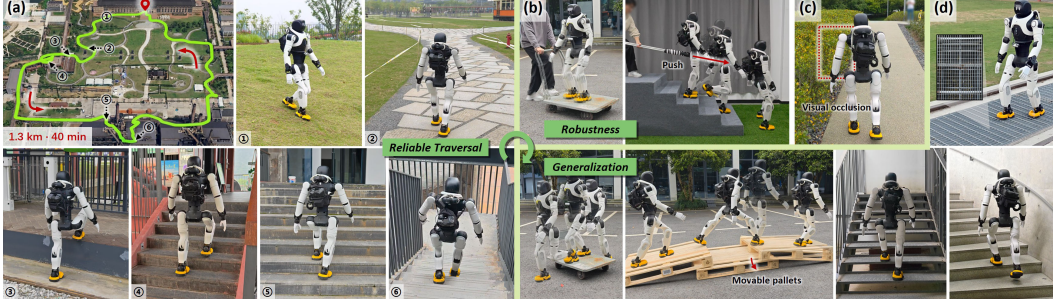


Figure 9: (a) The humanoid completes a 1.3 km, 40 min open-world traversal in an industrial heritage park, reliably crossing stairs of varying sizes, high platforms, rough ground, and grassy slopes. We further test robustness to (b) external disturbances, including wheeled-trolley shaking and pushes on descending stairs, and (c) visual occlusions from tall grass. (d) Zero-shot transfer to OOD terrains, including perforated grid flooring, a slippery trolley, movable forklift pallets, narrow 1.3 m-wide open-riser stairs, and spiral stairs.

Distribution (OOD) Generalization: despite no prior exposure to perforated grids, slippery trolley, or movable pallets during training, the policy can traverse them stably. On narrow open-riser stairs and spiral stairs, it handles visual interference from holes and handrails, avoids sidewalls that could trap the feet, and adapts to varying stair widths, showing excellent open-world generalization.

4.8 Cross-Platform Transfer

To assess cross-platform generalization, we deploy SSR on a full-size DEEP Robotics DR02 humanoid. Despite major differences from AgiBot X2, Appendix E shows successful transfer.

5 Conclusion

In this paper, we present SSR, a unified humanoid traversal framework that maps egocentric vision to stable, high-quality whole-body locomotion over diverse open-world terrains. Through predictive foothold reward, SSR guides foresighted foot placement during swing and acquires precise stepping in highly dynamic and complex settings. Our framework also leverages equivariant latent-space symmetry augmentation to efficiently learn bilateral coordination, while terrain-specific multi-discriminator further encourages consistently natural motion across terrains. Extensive experiments demonstrate that SSR generalizes surefooted, symmetric, and human-like traversal to a broad spectrum of challenging scenarios, supporting reliable long-horizon real-world deployment.

6 Limitations and Future Work

While SSR traverses diverse terrains effectively, several limitations remain. First, it relies on a fixed forward-facing depth camera, which limits perceptual coverage and mainly supports forward or diagonal obstacle crossing. Future work could combine multiple cameras or active view control for omnidirectional traversal. Second, SSR shows encouraging robustness to visual occlusion and some OOD cases outdoors, yet sim-to-real gap in depth sensing can still affects performance in extreme conditions. For example, on metal grid stairs under strong sunlight, specular reflections can create missing regions and distort local geometry estimation. Future efforts will investigate more realistic visual augmentations and richer RGB semantics to reduce this gap. Third, although SSR produces natural whole-body motion, it focuses on foot-ground contact, leaving room for upper-body interaction in more demanding skills like arm-assisted climbing or crawling through confined spaces. Future studies will explore multi-contact motion and human demonstrations to expand mobility.

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A Policy Training Details

A.1 Terrain Curriculum

As shown in Fig. 10, we generate five types of terrain for policy learning. Each terrain type contains 20 difficulty levels, with the difficulty progressively increased according to the curriculum strategy in [57]. Each terrain instance is constructed over a rectangular region measuring $8\text{ m} \times 4\text{ m}$.

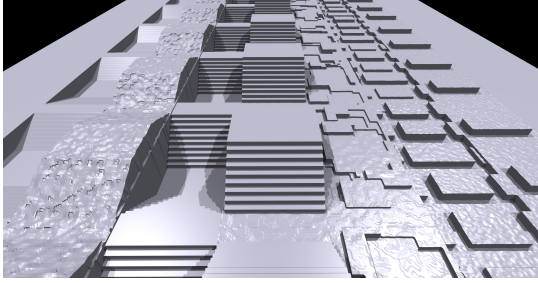


Figure 10: Overview of terrain types used for training.

Terrain Type	Parameter	Range
Slope	Incline	$[0, 20]^\circ$
Stairs	Step length	$[0.25, 0.4]\text{ m}$
	Step height	$[0.05, 0.2]\text{ m}$
Discrete	Obstacle height	$[0.05, 0.2]\text{ m}$
Gaps	Width	$[0.1, 0.8]\text{ m}$
	Depth	$[0.3, 0.6]\text{ m}$
Platforms	Height	$[0.05, 0.4]\text{ m}$

Table 3: Parameter ranges in the terrain curriculum.

A.2 Commands

At timestep t , the locomotion command is defined as $\mathbf{u}_t = [v_x^{\text{cmd}}, v_y^{\text{cmd}}, \omega_z^{\text{cmd}}]^\top$. During training, the robot is commanded to traverse the terrain along a straight path aligned with the x -axis of the world frame while tracking a target heading. The yaw angular velocity command ω_z^{cmd} is computed from the heading error, while the translational commands v_x^{cmd} and v_y^{cmd} are obtained by projecting the prescribed forward velocity from the world frame into the base frame. The terrain-specific ranges of the forward velocity and target heading are summarized in Table 4.

Terrain Type	Forward Velocity (m/s)	Heading ($^\circ$)
Slope	$\mathcal{U}(-1, 1)$	$\mathcal{U}(-80, 80)$
Stairs, discrete	$\mathcal{U}(0, 1.2)$	$\mathcal{U}(-30, 30)$
Gap, high platform	$\mathcal{U}(0, 1.2)$	$\mathcal{U}(-15, 15)$

Table 4: Ranges of the forward velocity and target heading.

To encourage omnidirectional locomotion, we further uniformly sample $\mathbf{u}_t$ in the base frame from a predefined command space after the robot has cleared the highest terrain difficulty and exited the terrain region. This stage exposes the policy to diverse combinations of forward, lateral, and turning motions beyond structured terrain-traversal commands. The corresponding command ranges are listed in Table 5.

Command Term	Range
Forward velocity	$\mathcal{U}(-1, 1)\text{ m/s}$
Lateral velocity	$\mathcal{U}(-0.4, 0.4)\text{ m/s}$
Yaw angular velocity	$\mathcal{U}(-1.2, 1.2)\text{ rad/s}$

Table 5: Ranges of the locomotion commands.

A.3 Rewards

Table 6 summarizes the reward terms used for policy learning, grouped by reward category. For each term, the table provides its mathematical expression and coefficient. It also reports the advantage weight associated with each reward group in the multi-critic formulation. Table 7 defines the main symbols used in these reward expressions.

Term	Expression	Weight
(a) <i>Locomotion Reward</i>	r^l	$w_l = 1.0$
Linear Velocity Tracking	$\exp(-\ \mathbf{v}_{xy} - \mathbf{v}_{xy}^{\text{cmd}}\ _2^2 / \sigma_v)$	1.0
Angular Velocity Tracking	$\exp(-(\boldsymbol{\omega}_{\text{yaw}} - \boldsymbol{\omega}_{\text{yaw}}^{\text{cmd}})^2 / \sigma_\omega)$	0.8
Orientation	$\exp(-\ \mathbf{g}_{xy}\ ^2 / \sigma_g)$	0.5
Angular Velocity (xy)	$\exp(-\ \boldsymbol{\omega}_{xy}\ _2^2 / \sigma_\omega)$	0.25
Base Height	$\exp(-(h_b - h_b^{\text{tar}})^2 / \sigma_h)$	0.4
Action Rate	$\frac{1}{N} \ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.12
Smoothness	$\frac{1}{N} \ \mathbf{a}_t - 2\mathbf{a}_{t-1} + \mathbf{a}_{t-2}\ _2^2$	-0.06
DoF Velocity	$\frac{1}{N} \ \dot{\boldsymbol{\theta}} / \dot{\boldsymbol{\theta}}^{\text{max}}\ _2^2$	-0.96
DoF Torque	$\frac{1}{N} \ \boldsymbol{\tau} / \boldsymbol{\tau}^{\text{max}}\ _2^2$	-0.6
DoF Deviation	$\frac{1}{N} \ \boldsymbol{\theta} - \boldsymbol{\theta}^{\text{def}}\ _1$	-1.8
DoF Position Limits	$\frac{1}{N} \ \mathbb{1}(\boldsymbol{\theta} \notin [\boldsymbol{\theta}^{\text{min}}, \boldsymbol{\theta}^{\text{max}}])\ _1$	-1.5
DoF Velocity Limits	$\frac{1}{N} \ \mathbb{1}(\dot{\boldsymbol{\theta}} > \dot{\boldsymbol{\theta}}^{\text{max}})\ _1$	-6.0
DoF Torque Limits	$\frac{1}{N} \ \mathbb{1}(\boldsymbol{\tau} > \boldsymbol{\tau}^{\text{max}})\ _1$	-6.0
Stand Still	$\frac{1}{N} \ \boldsymbol{\theta} - \boldsymbol{\theta}^{\text{def}}\ _1 \cdot \mathbb{1}(\ \mathbf{u}_t\ _2 \leq \epsilon)$	-0.12
Single Support	$\mathbb{1}[\ \mathbf{u}_t\ _2 \leq \epsilon \vee \exists t' \in [t - 0.2\text{s}, t] \text{ s.t. } n_{t'}^{\text{con}} = 1]$	0.2
Impact Velocity	$\sum_{i=1}^2 v_{z,i}^f \cdot c_i$	-1.3
Contact Slippage	$\sum_{i=1}^2 \ \mathbf{v}_{xy,i}^f\ _2^2 \cdot c_i$	-0.2
Feet Air Time	$\sum_{i=1}^2 \max(t_{\text{air},i} - t_{\text{air}}^{\text{tar}}, 0) \cdot c_i^{\text{first}}$	-2.0
Feet Stumble	$\sum_{i=1}^2 \mathbb{1}(\ \mathbf{F}_{xy,i}^f\ _2 > 3 F_{z,i}^f)$	-2.0
Feet Lateral Distance	$\exp(\min(d_y - d_y^{\text{min}}, 0) / \sigma_d)$	0.08
(b) <i>Foothold Reward</i>	r^f	$w_f = 0.25$
Foothold Safety	$\exp(-(\sum_{i=1}^2 \tilde{\rho}_{i,t})^2 / \sigma_f)$	1.0
(c) <i>Style Reward</i>	r^s	$w_s = 0.2$
AMP	$\max[0, 1 - 0.25(D_i(\boldsymbol{\psi}_t) - 1)^2]$	1.0

Table 6: Reward terms and weights.

Symbol	Description
$\sigma_v, \sigma_\omega, \sigma_g, \sigma_h, \sigma_d, \sigma_f$	Gaussian variance scales, set to 0.25, 0.25, 0.01, 0.01, 0.03, and 0.0625.
$\boldsymbol{\theta}^{\text{min}}, \boldsymbol{\theta}^{\text{max}}$	Lower and upper joint position limits.
$\boldsymbol{\theta}^{\text{def}}$	Default joint positions.
$\dot{\boldsymbol{\theta}}^{\text{max}}$	Maximum allowable joint velocities.
$\boldsymbol{\tau}$	Computed joint torques.
$\boldsymbol{\tau}^{\text{max}}$	Maximum allowable joint torques.
ϵ	Threshold for identifying zero-command, set to 0.15.
h_b, h_b^{tar}	Base height and target base height relative to the ground. $h_b^{\text{tar}} = 0.68$.
$t_{\text{air},i}, t_{\text{air}}^{\text{tar}}$	Air time of foot i and target air time. $t_{\text{air}}^{\text{tar}} = 0.4$.
c_i^{first}	Binary indicator of whether foot i makes its first contact with the ground.
$n_{t'}^{\text{con}}$	Number of feet in contact with the ground at time t' .
$\mathbf{F}_i^f$	Contact force on foot i .
$\mathbf{v}_i^f$	Velocity of foot i .
d_y	Lateral distance between the two feet.
d_y^{min}	Minimum allowable lateral distance between two feet, set to 0.22.

Table 7: Symbols in the reward definitions.

A.4 Network Architectures

Table 8 summarizes the network architectures used in our framework. The encoder architecture is further illustrated in Fig. 11.

Module	Submodule	Component	Structure	
Asymmetric Actor-Critic	MoE Actor	Number of experts	5	
		Expert MLP hidden size	[1024, 512, 128]	
		Gate MLP hidden size	128	
		Activation	ELU	
	Multi-Critic	Number of critics	3	
		MLP size	[512, 256, 128]	
		Activation	ELU	
Encoder	Proprioception Encoder	MLP hidden size	[512, 256, 128]	
		Activation	ELU	
		MLP embedding dim	128	
	Depth Encoder	CNN Channels	[32, 64, 128]	
		CNN kernel sizes	[8, 4, 3]	
		CNN strides	[4, 2, 2]	
		Activation	ELU	
		CNN embedding dim	128	
		Temporal Encoder	GRU hidden dim	256
			GRU layers	1
	Fusion Encoder	MLP hidden size	[512, 256, 128]	
		Activation	ELU	
MLP embedding dim		51		
	Latent Linear Heads	Output dims	$\hat{\mathbf{v}}_t : 3, \hat{\mathbf{z}}_t^f : 16, \hat{\mathbf{z}}_t^b : 16, \hat{\mathbf{z}}_t^p : 16$	
Next-Foothold Predictor		MLP size	[256, 128]	
		Activation	ELU	
Multi-Discriminator		Number of discriminators	5	
		MLP hidden size	[512, 256, 128]	
		Activation	ReLU	

Table 8: Network architectures.

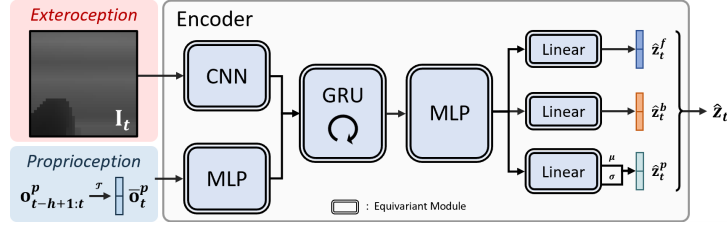


Figure 11: Encoder architecture. Double lines denote submodules built with equivariant layers, whereas single lines denote standard modules.

A.5 Training Hyperparameters

Component	Parameter	Value
PPO	Learning rate	5×10^{-4}
	Value loss coef	1.0
	Entropy coef	0.005
	Desired KL	0.01
	Minimum policy std	0.2
	GAE λ	0.95
	Reward discount γ	0.99
	Number of environments	4096
	Num steps per iteration	24
	Num learning epochs	5
Encoder	Num mini-batches	4
	Learning rate	5×10^{-4}
	Velocity MSE coef	2.0
	Base height map MSE coef	2.0
	Foot height maps MSE coef	1.0
	Next proprioception MSE coef	5.0
Next-Foothold Predictor	VAE KL divergence coef	1.0
	Learning rate	5×10^{-4}
Discriminator	Learning rate	1×10^{-4}
	Weight decay	1×10^{-4}
	Gradient penalty coef w^{GP}	20.0

Table 9: Training hyperparameters.

Table 9 summarizes the main hyperparameters used to train the components of our framework.

A.6 Domain Randomization

Table 10 summarizes the domain randomization settings used in our framework, including those applied to the system dynamics and depth sensing.

Category	Parameter	Range / Value
Dynamics	Base added mass	$\mathcal{U}(-2.0, 12.5)$ kg
	Base CoM offset	$\mathcal{U}(-0.1, 0.1)$ m
	Link mass	$\mathcal{U}(0.9, 1.1) \times \text{default}$
	Link CoM offset	$\mathcal{U}(-0.01, 0.01)$ m
	Motor strength	$\mathcal{U}(0.8, 1.2) \times \text{default}$
	PD gains	$\mathcal{U}(0.9, 1.1) \times \text{default}$
	Ground friction	$\mathcal{U}(0.2, 1.25)$
	Action delay	$\mathcal{U}(0, 2)$ ms
	Push interval	12 s
Depth Sensing	Push velocity	$\mathcal{U}(-0.4, 0.4)$ m/s
	Camera position offset	$\mathcal{U}(-0.01, 0.01)$ m
	Camera orientation offset	$\mathcal{U}(-1.0, 1.0)$ rad
	Horizontal FoV offset	$\mathcal{U}(-1, 1)^\circ$
	Vertical FoV offset	$\mathcal{U}(-1, 1)^\circ$

Table 10: Domain randomization settings.

A.7 Termination Criteria

An episode is terminated when any of the following conditions is satisfied:

- The episode reaches its maximum duration.
- The robot moves outside the terrain boundary.
- The contact force on any termination-contact link exceeds a predefined threshold.
- The robot exhibits severe orientation instability.
- On gap terrain, any foot drops below a predefined threshold beneath the ground plane.

A.8 Motion Prior Dataset

For each terrain type, the corresponding motion priors are sourced from two datasets: customized motion recordings collected using a motion capture system and a selected subset of the open-source AMASS dataset [58]. All motion sequences are retargeted to our humanoid following Xie et al. [59]. Table 11 summarizes the statistics of the motion datasets used for training.

Terrain Type	Flat-ground motions			Terrain traversals					Total Time (s)
	Forward	Backward	Turn	Slope	Stairs	Step	Gap	Platform	
Slope	✓	✓	✓	✓					261.3
Stairs	✓		✓		✓				295.8
Discrete	✓		✓			✓			136.0
Gap	✓		✓				✓		174.5
Platform	✓		✓			✓		✓	181.6

Table 11: Motion dataset statistics.

B Equivariant Network Details

B.1 Mirror Transformations and Symmetry-Structured Encoder Inputs

This section formalizes the mirror transformations and symmetry operators used in our method. We first define the mirror transformations used for data augmentation in Sec. 3.3. We then describe the structure of the encoder inputs and introduce the associated operators for constructing symmetry-structured representations, which are also used in the proofs in Appendix B.2.

We begin with the mirror transformation $\mathcal{M}_c$ for the implicit latent representation $\hat{\mathbf{z}}_t$, defined as an exchange between the left and right channel groups. More generally, $\mathcal{M}_c$ is the primitive mirror operator for symmetry-structured vectors. For a vector whose channel dimension is partitioned into two paired groups, $\mathcal{M}_c$ swaps the two groups:

$$\mathbf{X} = [\mathbf{X}^{(L)}, \mathbf{X}^{(R)}]^\top \quad \mathcal{M}_c(\mathbf{X}) = [\mathbf{X}^{(R)}, \mathbf{X}^{(L)}]^\top. \quad (4)$$

We next define the mirror transformation $\mathcal{M}_p$ for proprioception. To specify its action on joint-space variables, we introduce the following block notation. For any joint-space quantity, including joint positions, joint velocities, and actions, we write $\mathbf{x} = [\mathbf{x}_R^{\text{leg}}, \mathbf{x}_L^{\text{leg}}, x^{\text{waist}}, \mathbf{x}_R^{\text{arm}}, \mathbf{x}_L^{\text{arm}}]^\top \in \mathbb{R}^{21}$. Each leg block is ordered as (*hip pitch, hip roll, hip yaw, knee, ankle pitch, ankle roll*), and each arm block is ordered as (*shoulder pitch, shoulder roll, shoulder yaw, elbow*).

Under reflection about the sagittal plane, the left and right kinematic chains are exchanged, and the sign of each coordinate depends on the corresponding physical axis convention. We therefore define the sign vectors for the joint blocks as $\mathbf{s}_{\text{leg}} = [1, -1, -1, 1, 1, -1]^\top$ and $\mathbf{s}_{\text{arm}} = [1, -1, -1, 1]^\top$. The resulting joint-block reflection is defined as

$$\mathcal{F}(\mathbf{x}) = [\mathbf{x}_L^{\text{leg}} \odot \mathbf{s}_{\text{leg}}, \mathbf{x}_R^{\text{leg}} \odot \mathbf{s}_{\text{leg}}, -x^{\text{waist}}, \mathbf{x}_L^{\text{arm}} \odot \mathbf{s}_{\text{arm}}, \mathbf{x}_R^{\text{arm}} \odot \mathbf{s}_{\text{arm}}]^\top, \quad (5)$$

where $\odot$ denotes element-wise multiplication. For the remaining proprioceptive quantities with explicit physical semantics, namely base-frame polar vectors, base angular velocities, and velocity commands, we use the sign vectors $\mathbf{s}_y = [1, -1, 1]^\top$, $\mathbf{s}_\omega = [-1, 1, -1]^\top$, and $\mathbf{s}_u = [1, -1, -1]^\top$.

Table 12 summarizes the mirror transformations used for data augmentation in Sec. 3.3, including those for proprioception, latent representations, actor observations, privileged states, and actions.

Transformation	Component	Variable	Mirror rule	Dim.
Proprioception $\mathbf{o}_t^p$				
$\mathcal{M}_p$	Base angular velocity	$\boldsymbol{\omega}_t$	$\boldsymbol{\omega}_t \odot \mathbf{s}_\omega$	3
	Gravity vector	$\mathbf{g}_t$	$\mathbf{g}_t \odot \mathbf{s}_y$	3
	Velocity command	$\mathbf{u}_t$	$\mathbf{u}_t \odot \mathbf{s}_u$	3
	Joint angles	$\boldsymbol{\theta}_t$	$\mathcal{F}(\boldsymbol{\theta}_t)$	21
	Joint angular velocities	$\dot{\boldsymbol{\theta}}_t$	$\mathcal{F}(\dot{\boldsymbol{\theta}}_t)$	21
	Previous action	$\mathbf{a}_{t-1}$	$\mathcal{F}(\mathbf{a}_{t-1})$	21
Encoder latent $\mathbf{l}_t$				
$\mathcal{M}_l$	Estimated base velocity	$\hat{\mathbf{v}}_t$	$\mathcal{M}_v(\hat{\mathbf{v}}_t) = \hat{\mathbf{v}}_t \odot \mathbf{s}_y$	3
	Foot height maps latent	$\hat{\mathbf{z}}_t^f$	$\mathcal{M}_c(\hat{\mathbf{z}}_t^f)$	16
	Base height map latent	$\hat{\mathbf{z}}_t^b$	$\mathcal{M}_c(\hat{\mathbf{z}}_t^b)$	16
	Next-proprioception latent	$\hat{\mathbf{z}}_t^p$	$\mathcal{M}_c(\hat{\mathbf{z}}_t^p)$	16
Actor observation $\mathbf{o}_t^a$				
$\mathcal{M}_o$	Proprioception	$\mathbf{o}_t^p$	$\mathcal{M}_p(\mathbf{o}_t^p)$	72
	Encoder latent	$\mathbf{l}_t$	$\mathcal{M}_l(\mathbf{l}_t)$	51
Privileged state $\mathbf{s}_t$				
$\mathcal{M}_s$	Proprioceptive observation	$\mathbf{o}_t^p$	$\mathcal{M}_p(\mathbf{o}_t^p)$	72
	Base linear velocity	$\mathbf{v}_t$	$\mathbf{v}_t \odot \mathbf{s}_y$	3
	Foot linear velocities	$[\mathbf{v}_{t,R}^f, \mathbf{v}_{t,L}^f]^\top$	$[\mathbf{v}_{t,L}^f \odot \mathbf{s}_y, \mathbf{v}_{t,R}^f \odot \mathbf{s}_y]^\top$	6
	Foot contact	$[c_{t,R}, c_{t,L}]^\top$	$[c_{t,L}, c_{t,R}]^\top$	2
	Hand positions	$[\mathbf{p}_{t,R}^h, \mathbf{p}_{t,L}^h]^\top$	$[\mathbf{p}_{t,L}^h \odot \mathbf{s}_y, \mathbf{p}_{t,R}^h \odot \mathbf{s}_y]^\top$	6
	Foot positions	$[\mathbf{p}_{t,R}^f, \mathbf{p}_{t,L}^f]^\top$	$[\mathbf{p}_{t,L}^f \odot \mathbf{s}_y, \mathbf{p}_{t,R}^f \odot \mathbf{s}_y]^\top$	6
	Base height map	$\mathbf{H}_t^b$	$\mathcal{M}_{2D}(\mathbf{H}_t^b)$	18×10
	Foot height maps	$(\mathbf{H}_{t,R}^f, \mathbf{H}_{t,L}^f)$	$(\mathcal{M}_{2D}(\mathbf{H}_{t,L}^f), \mathcal{M}_{2D}(\mathbf{H}_{t,R}^f))$	$2 \times 10 \times 5$
Action $\mathbf{a}_t$				
$\mathcal{M}_a$	Action	$\mathbf{a}_t$	$\mathcal{F}(\mathbf{a}_t)$	21

Table 12: Component-wise mirror rules used in the data augmentation procedure for proprioceptions, latent representations, actor observations, privileged states, and actions.

Next, we analyze the structure of the encoder inputs and define the corresponding mirror operators used in the proofs in Appendix B.2. Because mirror equivariance is implemented through modules with explicitly paired left-right channels, their inputs must share the same symmetry-structured form. In our setting, this form is obtained by organizing the features into paired left and right channels. We describe this construction below for temporal proprioceptive and image inputs.

We first consider the temporal proprioception $\mathbf{o}_{t-h+1:t}^p$. This representation does not explicitly separate the left and right sides. We therefore introduce a fixed reorganization operator $\mathcal{T}$ that maps the length- h proprioceptive history into a symmetry-structured left-right representation:

$$\bar{\mathbf{o}}_t^p = \mathcal{T}(\mathbf{o}_{t-h+1:t}^p) = [\bar{\mathbf{o}}_{t,L}^p, \bar{\mathbf{o}}_{t,R}^p]^\top, \quad \bar{\mathbf{o}}_{t,L}^p, \bar{\mathbf{o}}_{t,R}^p \in \mathbb{R}^{42 \times h}. \quad (6)$$

Each branch is constructed by concatenating the corresponding per-frame slices over the past h time steps. The definitions of the per-frame slices $\bar{\mathbf{o}}_{t-k,L}^p$ and $\bar{\mathbf{o}}_{t-k,R}^p$, for $k \in \{0, \dots, h-1\}$, are listed in Table 13. Under this reorganization, the mirror transformation reduces to simple branch exchange:

$$\mathcal{M}_c(\bar{\mathbf{o}}_t^p) = [\bar{\mathbf{o}}_{t,R}^p, \bar{\mathbf{o}}_{t,L}^p]^\top. \quad (7)$$

Component	Left slice $\bar{\mathbf{o}}_{t-k,L}^p$	Right slice $\bar{\mathbf{o}}_{t-k,R}^p$	Dim.
Base angular velocity	$\boldsymbol{\omega}_{t-k}$	$\boldsymbol{\omega}_{t-k} \odot \mathbf{s}_\omega$	3
Gravity vector	$\mathbf{g}_{t-k}$	$\mathbf{g}_{t-k} \odot \mathbf{s}_y$	3
Velocity command	$\mathbf{u}_{t-k}$	$\mathbf{u}_{t-k} \odot \mathbf{s}_u$	3
Joint angles	$[\theta_{t-k,L}^{\text{leg}}, \theta_{t-k,L}^{\text{arm}}, \theta_{t-k,L}^{\text{waist}}]^\top$	$[\theta_{t-k,R}^{\text{leg}} \odot \mathbf{s}_{\text{leg}}, \theta_{t-k,R}^{\text{arm}} \odot \mathbf{s}_{\text{arm}}, -\theta_{t-k,R}^{\text{waist}}]^\top$	11
Joint angular velocities	$[\dot{\theta}_{t-k,L}^{\text{leg}}, \dot{\theta}_{t-k,L}^{\text{arm}}, \dot{\theta}_{t-k,L}^{\text{waist}}]^\top$	$[\dot{\theta}_{t-k,R}^{\text{leg}} \odot \mathbf{s}_{\text{leg}}, \dot{\theta}_{t-k,R}^{\text{arm}} \odot \mathbf{s}_{\text{arm}}, -\dot{\theta}_{t-k,R}^{\text{waist}}]^\top$	11
Previous action	$[\mathbf{a}_{t-k-1,L}^{\text{leg}}, \mathbf{a}_{t-k-1,L}^{\text{arm}}, \mathbf{a}_{t-k-1,L}^{\text{waist}}]^\top$	$[\mathbf{a}_{t-k-1,R}^{\text{leg}} \odot \mathbf{s}_{\text{leg}}, \mathbf{a}_{t-k-1,R}^{\text{arm}} \odot \mathbf{s}_{\text{arm}}, -\mathbf{a}_{t-k-1,R}^{\text{waist}}]^\top$	11

Table 13: Per-frame left and right slices used by the reorganization operator $\mathcal{T}$.

We then consider the image input $\mathbf{I}_t$. Images naturally exhibits a left-right spatial structure. We therefore use the horizontal reflection operator $\mathcal{M}_{2D}$ for two-dimensional vectors. For $\mathbf{Y} \in \mathbb{R}^{H \times W}$, including an image or any image-derived representation, $\mathcal{M}_{2D}$ acts along the width dimension and performs a horizontal flip:

$$[\mathcal{M}_{2D}(\mathbf{Y})]_{i,j} = \mathbf{Y}_{i,W+1-j}, \quad i = 1, \dots, H, \quad j = 1, \dots, W. \quad (8)$$

Although an image already has a left-right spatial structure, it does not directly provide paired left-right channels along the channel dimension. To address this issue, we introduce a lift convolution layer in the equivariant CNN (Appendix B.2), which lifts the input image to a symmetry-structured feature representation with explicitly paired left and right channels.

B.2 Equivariance Proofs for Network Components

To construct an encoder that satisfies the desired equivariance property, we design each of its constituent modules as an equivariant network, as illustrated in Fig. 11. These modules include equivariant MLPs, equivariant GRUs, and an equivariant CNN, which can in turn be decomposed into compositions of equivariant linear and convolutional layers. In this section, we first present the construction and properties of these equivariant network components and provide the corresponding proofs. We then use these results to establish the equivariance of the complete encoder in Sec. 3.3.

Equivariant Linear Layer An equivariant linear layer has the same affine form as a standard linear layer, but its weight matrix and bias are constrained to follow a symmetry-preserving block structure. We define $f : \mathbb{R}^{2C_{\text{in}}} \rightarrow \mathbb{R}^{2C_{\text{out}}}$ as

$$f(\mathbf{x}) = W\mathbf{x} + \mathbf{b}, \quad \mathbf{x} = \begin{bmatrix} \mathbf{x}^{(1)} \\ \mathbf{x}^{(2)} \end{bmatrix}, \quad W = \begin{bmatrix} A & B \\ B & A \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} \mathbf{a} \\ \mathbf{a} \end{bmatrix}. \quad (9)$$

Here, $\mathbf{x} \in \mathbb{R}^{2C_{\text{in}}}$ is the symmetry-structured input feature vector, $A, B \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ are learnable weight blocks, and $\mathbf{a} \in \mathbb{R}^{C_{\text{out}}}$ is the shared bias.

Proposition B.1. *The linear layer f is $\mathcal{M}_c$ -equivariant:*

$$f(\mathcal{M}_c \mathbf{x}) = \mathcal{M}_c f(\mathbf{x}). \quad (10)$$

Proof. Direct substitution yields

$$f(\mathcal{M}_c \mathbf{x}) = \begin{bmatrix} A\mathbf{x}^{(2)} + B\mathbf{x}^{(1)} + \mathbf{a} \\ B\mathbf{x}^{(2)} + A\mathbf{x}^{(1)} + \mathbf{a} \end{bmatrix} = \mathcal{M}_c \begin{bmatrix} A\mathbf{x}^{(1)} + B\mathbf{x}^{(2)} + \mathbf{a} \\ B\mathbf{x}^{(1)} + A\mathbf{x}^{(2)} + \mathbf{a} \end{bmatrix} = \mathcal{M}_c f(\mathbf{x}).$$

Equivariant MLP The equivariant MLP follows the standard MLP form, except that each linear layer is replaced with an equivariant linear layer f_ℓ :

$$F_{\text{MLP}}(\mathbf{x}) = f_L \left(\sigma \left(f_{L-1} \left(\cdots \sigma \left(f_1(\mathbf{x}) \right) \cdots \right) \right) \right), \quad (11)$$

where σ denotes the activation function.

Proposition B.2. *The equivariant MLP F_{MLP} is $\mathcal{M}_c$ -equivariant:*

$$F_{\text{MLP}}(\mathcal{M}_c \mathbf{x}) = \mathcal{M}_c F_{\text{MLP}}(\mathbf{x}). \quad (12)$$

Proof. Let

$$G_\ell = \begin{cases} \sigma \circ f_\ell, & \ell = 1, \dots, L-1, \\ f_L, & \ell = L. \end{cases}$$

By Proposition B.1, each equivariant linear layer f_ℓ commutes with $\mathcal{M}_c$. Since σ is applied element-wise, it also commutes with $\mathcal{M}_c$. Hence every G_ℓ is $\mathcal{M}_c$ -equivariant. Because a composition of equivariant maps is equivariant,

$$F_{\text{MLP}} = G_L \circ G_{L-1} \circ \cdots \circ G_1$$

also satisfies

$$F_{\text{MLP}}(\mathcal{M}_c \mathbf{x}) = \mathcal{M}_c F_{\text{MLP}}(\mathbf{x}).$$

Equivariant GRU The equivariant GRU follows the standard update rule, with $\mathbf{h}_t = F_{\text{GRU}}(\mathbf{x}_t, \mathbf{h}_{t-1})$, where all affine transformations use the same block-structured parameterization as the equivariant linear layer:

$$\begin{cases} \mathbf{r}_t = \sigma(W_{\text{ir}}\mathbf{x}_t + \mathbf{b}_{\text{ir}} + W_{\text{hr}}\mathbf{h}_{t-1} + \mathbf{b}_{\text{hr}}), \\ \mathbf{z}_t = \sigma(W_{\text{iz}}\mathbf{x}_t + \mathbf{b}_{\text{iz}} + W_{\text{hz}}\mathbf{h}_{t-1} + \mathbf{b}_{\text{hz}}), \\ \mathbf{n}_t = \tanh(W_{\text{in}}\mathbf{x}_t + \mathbf{b}_{\text{in}} + \mathbf{r}_t \odot (W_{\text{hn}}\mathbf{h}_{t-1} + \mathbf{b}_{\text{hn}})), \\ \mathbf{h}_t = (\mathbf{1} - \mathbf{z}_t) \odot \mathbf{n}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1}. \end{cases} \quad (13)$$

Here, $\mathbf{x}_t \in \mathbb{R}^{2C_{\text{in}}}$ denotes the input feature at time t ; $\mathbf{h}_{t-1}$ and $\mathbf{h}_t$ are the previous and updated hidden states in $\mathbb{R}^{2C_{\text{h}}}$, and $\mathbf{r}_t$, $\mathbf{z}_t$, and $\mathbf{n}_t$ denote the reset gate, update gate, and candidate hidden state, respectively.

Proposition B.3. *The one-step GRU update is equivariant under channel exchange:*

$$F_{\text{GRU}}(\mathcal{M}_c \mathbf{x}_t, \mathcal{M}_c \mathbf{h}_{t-1}) = \mathcal{M}_c F_{\text{GRU}}(\mathbf{x}_t, \mathbf{h}_{t-1}). \quad (14)$$

Proof. By Proposition B.1, all affine maps in the GRU update commute with $\mathcal{M}_c$. Since σ , $\tanh$, subtraction from $\mathbf{1}$, and the Hadamard product are all pointwise operations, they also commute with $\mathcal{M}_c$. Therefore

$$\mathbf{r}'_t = \mathcal{M}_c \mathbf{r}_t, \quad \mathbf{z}'_t = \mathcal{M}_c \mathbf{z}_t.$$

For the candidate hidden state,

$$\begin{aligned} \mathbf{n}'_t &= \tanh \left(W_{\text{in}} \mathcal{M}_c \mathbf{x}_t + \mathbf{b}_{\text{in}} + \mathbf{r}'_t \odot (W_{\text{hn}} \mathcal{M}_c \mathbf{h}_{t-1} + \mathbf{b}_{\text{hn}}) \right) \\ &= \tanh \left(\mathcal{M}_c (W_{\text{in}} \mathbf{x}_t + \mathbf{b}_{\text{in}} + \mathbf{r}_t \odot (W_{\text{hn}} \mathbf{h}_{t-1} + \mathbf{b}_{\text{hn}})) \right) = \mathcal{M}_c \mathbf{n}_t. \end{aligned}$$

Hence

$$\begin{aligned} F_{\text{GRU}}(\mathcal{M}_c \mathbf{x}_t, \mathcal{M}_c \mathbf{h}_{t-1}) &= (\mathbf{1} - \mathbf{z}'_t) \odot \mathbf{n}'_t + \mathbf{z}'_t \odot \mathcal{M}_c \mathbf{h}_{t-1} \\ &= \mathcal{M}_c ((\mathbf{1} - \mathbf{z}_t) \odot \mathbf{n}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1}) = \mathcal{M}_c F_{\text{GRU}}(\mathbf{x}_t, \mathbf{h}_{t-1}). \end{aligned}$$

Equivariant Convolution Layer We define the equivariant convolutional layers using cross-correlation and symmetry-constrained kernels. For an input feature map $\mathbf{x} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, let $\bar{\mathbf{x}} \in \mathbb{R}^{C_{\text{in}} \times \bar{H} \times \bar{W}}$ denote its padded version, where $\bar{H} = H + 2p_H$ and $\bar{W} = W + 2p_W$. Given a convolution kernel $\mathbf{k} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K_H \times K_W}$, we define cross-correlation with stride s as

$$(\mathbf{x} \star \mathbf{k})_{o,h,w} = \sum_{i=1}^{C_{\text{in}}} \sum_{u=1}^{K_H} \sum_{v=1}^{K_W} \mathbf{k}_{o,i,u,v} \bar{\mathbf{x}}_{i,(h-1)s+u,(w-1)s+v}. \quad (15)$$

Here, $h=1, \dots, H_o$ and $w=1, \dots, W_o$, where $H_o = \lfloor (\bar{H} - K_H)/s \rfloor + 1$ and $W_o = \lfloor (\bar{W} - K_W)/s \rfloor + 1$. When $s > 1$, we assume $\bar{W} - K_W$ is divisible by s so that horizontal sampling grid remains aligned under flipping. To describe mirror equivariance in the spatial domain, we define the horizontal flip operator on padded feature maps and kernels as

$$[\mathcal{M}_{2D}\bar{\mathbf{x}}]_{i,h,w} = \bar{\mathbf{x}}_{i,h,\bar{W}+1-w}, \quad [\mathcal{M}_{2D}\mathbf{k}]_{o,i,u,v} = \mathbf{k}_{o,i,u,K_W+1-v}. \quad (16)$$

An ordinary image does not explicitly contain paired symmetry channels. We therefore use a lift convolution to map it to a symmetry-structured representation:

$$f_1(\mathbf{x}) = \mathbf{x} \star K_1 + \mathbf{b}_1, \quad K_1 = \begin{bmatrix} \mathbf{k}_1 \\ \mathcal{M}_{2D}\mathbf{k}_1 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_1 \end{bmatrix}. \quad (17)$$

Here, $\mathbf{k}_1 \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K_H \times K_W}$ is a learnable kernel, and $\mathbf{a}_1 \in \mathbb{R}^{C_{\text{out}}}$ is a shared bias. After lifting, the subsequent layers preserve the symmetry-structured representation through constrained kernels. Each non-lift convolution is defined as

$$f_{\text{nl}}(\mathbf{x}) = \mathbf{x} \star K_{\text{nl}} + \mathbf{b}_{\text{nl}}, \quad \mathbf{x} = \begin{bmatrix} \mathbf{x}^{(1)} \\ \mathbf{x}^{(2)} \end{bmatrix}, \quad K_{\text{nl}} = \begin{bmatrix} \mathbf{k}_a & \mathbf{k}_b \\ \mathcal{M}_{2D}\mathbf{k}_b & \mathcal{M}_{2D}\mathbf{k}_a \end{bmatrix}, \quad \mathbf{b}_{\text{nl}} = \begin{bmatrix} \mathbf{a}_{\text{nl}} \\ \mathbf{a}_{\text{nl}} \end{bmatrix}. \quad (18)$$

Here, $\mathbf{x}^{(1)}, \mathbf{x}^{(2)} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ are the two symmetry branches, $\mathbf{k}_a, \mathbf{k}_b \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K_H \times K_W}$ are learnable kernels, and $\mathbf{a}_{\text{nl}} \in \mathbb{R}^{C_{\text{out}}}$ is a shared bias.

Lemma B.4. *Cross-correlation satisfies*

$$(\mathcal{M}_{2D}\mathbf{x}) \star \mathbf{k} = \mathcal{M}_{2D}(\mathbf{x} \star (\mathcal{M}_{2D}\mathbf{k})). \quad (19)$$

Proof. For the left-hand side,

$$[(\mathcal{M}_{2D}\mathbf{x}) \star \mathbf{k}]_{o,h,w} = \sum_{i,u,v} \mathbf{k}_{o,i,u,v} \bar{\mathbf{x}}_{i,(h-1)s+u,\bar{W}+1-(w-1)s-v}.$$

For the right-hand side, substituting $v' = K_W + 1 - v$ gives

$$[\mathcal{M}_{2D}(\mathbf{x} \star (\mathcal{M}_{2D}\mathbf{k}))]_{o,h,w} = \sum_{i,u,v'} \mathbf{k}_{o,i,u,v'} \bar{\mathbf{x}}_{i,(h-1)s+u,(W_o-w)s+K_W+1-v'}.$$

Using $W_o = \frac{\bar{W}-K_W}{s} + 1$ and the alignment assumption, the last index equals

$$\bar{W} + 1 - (w-1)s - v',$$

so the two expressions coincide.

Proposition B.5. *The lift convolution satisfies*

$$f_1(\mathcal{M}_{2D}\mathbf{x}) = \mathcal{M}_c \mathcal{M}_{2D}(f_1(\mathbf{x})). \quad (20)$$

Proof. By Lemma B.4,

$$(\mathcal{M}_{2D}\mathbf{x}) \star \mathbf{k}_1 = \mathcal{M}_{2D}(\mathbf{x} \star (\mathcal{M}_{2D}\mathbf{k}_1)), \quad (\mathcal{M}_{2D}\mathbf{x}) \star (\mathcal{M}_{2D}\mathbf{k}_1) = \mathcal{M}_{2D}(\mathbf{x} \star \mathbf{k}_1).$$

Hence

$$f_1(\mathcal{M}_{2D}\mathbf{x}) = \begin{bmatrix} \mathcal{M}_{2D}(\mathbf{x} \star (\mathcal{M}_{2D}\mathbf{k}_1) + \mathbf{a}_1) \\ \mathcal{M}_{2D}(\mathbf{x} \star \mathbf{k}_1 + \mathbf{a}_1) \end{bmatrix} = \mathcal{M}_c \mathcal{M}_{2D} f_1(\mathbf{x}).$$

Proposition B.6. *The non-lift convolution satisfies*

$$f_{\text{nl}}(\mathcal{M}_c \mathcal{M}_{2\text{D}} \mathbf{x}) = \mathcal{M}_c \mathcal{M}_{2\text{D}} f_{\text{nl}}(\mathbf{x}). \quad (21)$$

Proof. Let

$$y_1 = \mathbf{x}^{(1)} \star \mathbf{k}_a + \mathbf{x}^{(2)} \star \mathbf{k}_b + \mathbf{a}_{\text{nl}}, \quad y_2 = \mathbf{x}^{(1)} \star (\mathcal{M}_{2\text{D}} \mathbf{k}_b) + \mathbf{x}^{(2)} \star (\mathcal{M}_{2\text{D}} \mathbf{k}_a) + \mathbf{a}_{\text{nl}}.$$

Then

$$f_{\text{nl}}(\mathbf{x}) = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Using Lemma B.4,

$$f_{\text{nl}}(\mathcal{M}_c \mathcal{M}_{2\text{D}} \mathbf{x}) = \begin{bmatrix} \mathcal{M}_{2\text{D}} y_2 \\ \mathcal{M}_{2\text{D}} y_1 \end{bmatrix} = \mathcal{M}_c \mathcal{M}_{2\text{D}} f_{\text{nl}}(\mathbf{x}).$$

Equivariant CNN The equivariant CNN is constructed by composing the layer-wise equivariant convolutions defined above. It consists of one lift convolution f_1 followed by $N - 1$ non-lift convolutions f_{nl}^i :

$$F_{\text{CNN}}(\mathbf{x}) = f_{\text{nl}}^{N-1} \left(\sigma \left(f_{\text{nl}}^{N-2} \left(\cdots \sigma \left(f_{\text{nl}}^1 \left(\sigma \left(f_1(\mathbf{x}) \right) \right) \right) \right) \right) \right). \quad (22)$$

Let $F_{\text{CNN}}(\mathbf{x}) \in \mathbb{R}^{2C_N \times H_N \times W_N}$ denote the output feature map of the equivariant CNN, where H_N and W_N are its spatial height and width.

Proposition B.7. *The equivariant CNN satisfies*

$$F_{\text{CNN}}(\mathcal{M}_{2\text{D}} \mathbf{x}) = \mathcal{M}_c \mathcal{M}_{2\text{D}} F_{\text{CNN}}(\mathbf{x}). \quad (23)$$

Proof. Let

$$T := \mathcal{M}_c \mathcal{M}_{2\text{D}}.$$

By Proposition B.5, the lift convolution is T -equivariant. By Proposition B.6, each non-lift convolution is also T -equivariant. Since σ is applied element-wise, it commutes with both channel exchange and horizontal flipping, and hence with T . Therefore every layer in the composition defining F_{CNN} is T -equivariant. By closure under composition,

$$F_{\text{CNN}}(\mathcal{M}_{2\text{D}} \mathbf{x}) = T F_{\text{CNN}}(\mathbf{x}) = \mathcal{M}_c \mathcal{M}_{2\text{D}} F_{\text{CNN}}(\mathbf{x}).$$

Corollary B.8. *If the output feature map has spatial resolution 1×1 , then*

$$F_{\text{CNN}}(\mathcal{M}_{2\text{D}} \mathbf{x}) = \mathcal{M}_c F_{\text{CNN}}(\mathbf{x}). \quad (24)$$

Theorem B.9. *Let E_ϕ denote the complete encoder used in Sec. 3.3, which takes the symmetry-structured proprioceptive input $\bar{\mathbf{o}}_t^p$ and the image input $\mathbf{I}_t$ and outputs the latent representation. Then the encoder is equivariant under the joint action of channel exchange on proprioception and horizontal reflection on images:*

$$E_\phi(\mathcal{M}_c \bar{\mathbf{o}}_t^p, \mathcal{M}_{2\text{D}} \mathbf{I}_t) = \mathcal{M}_l E_\phi(\bar{\mathbf{o}}_t^p, \mathbf{I}_t), \quad (25)$$

where $\mathcal{M}_l$ denotes the mirror operator acting on the latent representation.

Proof. The encoder in Sec. 3.3 is constructed by composing an equivariant proprioceptive branch, an equivariant visual branch, and a latent mapping defined on the fused symmetry-structured features.

First, proprioceptive branch is implemented using the equivariant modules established above. By Proposition B.3, the GRU block is $\mathcal{M}_c$ -equivariant. By Proposition B.2, each equivariant MLP used in the proprioceptive branch is also $\mathcal{M}_c$ -equivariant. Therefore, by closure under composition, the entire proprioceptive branch is $\mathcal{M}_c$ -equivariant.

Second, visual branch is implemented by equivariant CNN. By Proposition B.7, the CNN satisfies

$$F_{\text{CNN}}(\mathcal{M}_{2\text{D}}\mathbf{x}) = \mathcal{M}_c \mathcal{M}_{2\text{D}} F_{\text{CNN}}(\mathbf{x}).$$

For the architecture used in this work, the output spatial resolution is 1×1 . Hence, by Corollary B.8, the visual feature produced by the CNN satisfies

$$F_{\text{CNN}}(\mathcal{M}_{2\text{D}}\mathbf{I}_t) = \mathcal{M}_c F_{\text{CNN}}(\mathbf{I}_t).$$

Next, let $\oplus$ denote the feature fusion operation used by the encoder to combine the proprioceptive and visual branch outputs along the symmetry-structured channel dimension. Since both branch features transform under the same channel-exchange operator, this fusion preserves equivariance:

$$(\mathcal{M}_c \mathbf{u}) \oplus (\mathcal{M}_c \mathbf{v}) = \mathcal{M}_c (\mathbf{u} \oplus \mathbf{v}).$$

Finally, the fused feature is mapped to the latent representation by an equivariant MLP. By Proposition B.2, this mapping is equivariant with respect to the channel-exchange action on its input and the corresponding latent mirror operator $\mathcal{M}_l$ on its output. Combining the equivariance of the proprioceptive branch, the visual branch, the fusion step, and the final latent mapping yields

$$E_\phi(\mathcal{M}_c \bar{\mathbf{o}}_t^p, \mathcal{M}_{2\text{D}}\mathbf{I}_t) = \mathcal{M}_l E_\phi(\bar{\mathbf{o}}_t^p, \mathbf{I}_t).$$

This completes the proof.

C Efficient Depth Rendering with Self-Occlusion

To synthesize depth observations consistent with deployment-time sensing, we ray-cast against both the static terrain mesh and the robot’s dynamic link meshes. A naive implementation would update dynamic mesh poses and repeatedly refit the corresponding ray-query acceleration structures, such as bounding volume hierarchies (BVHs), at every simulation step, which quickly becomes a bottleneck in massively parallel training. To avoid this overhead, we keep each mesh in its local frame and instead transform each camera ray from the world frame into the queried mesh frame before ray-mesh intersection. Because these transformations are rigid, hit distances remain directly comparable across meshes. This yields an efficient depth renderer with self-occlusion, implemented in NVIDIA Warp [54], for large-scale GPU-parallel simulation. The procedure is summarized in Algorithm 1.

Algorithm 1: Efficient Depth Rendering with Self-Occlusion

Require: terrain mesh $\mathcal{M}^{\text{ter}}$ with pose $(\mathbf{p}^{\text{ter}}, \mathbf{R}^{\text{ter}})$; per-link meshes $\{\mathcal{M}_j^\ell\}_{j=1}^{N^\ell}$ with per-env link poses $\{(\mathbf{p}_{j,e}^\ell, \mathbf{R}_{j,e}^\ell)\}$; per-env camera poses $\{(\mathbf{p}_e^{\text{cam}}, \mathbf{R}_e^{\text{cam}})\}_{e=1}^{N^{\text{env}}}$; half-FoV tangents $(\tan_{x,e}, \tan_{y,e})$; depth bounds $(d_{\min}, d_{\max})$.

Ensure: clipped camera-depth maps $\mathbf{D} \in \mathbb{R}^{N^{\text{env}} \times H \times W}$.

▷ *Parallel mesh-based occluder rendering with one thread per pixel*

- 1: **for each** pixel $(e, h, w) \in [N^{\text{env}}] \times [H] \times [W]$ **in parallel do**
- 2: Image-plane ray coordinates: $t_x \leftarrow (2(w - \frac{1}{2})/W - 1) \tan_{x,e}, t_y \leftarrow (2(h - \frac{1}{2})/H - 1) \tan_{y,e}$
- 3: World frame ray origin: $\mathbf{o} \leftarrow \mathbf{p}_e^{\text{cam}}$
- 4: World frame ray direction: $\mathbf{d} \leftarrow \text{Normalize}(\text{Rot}(\mathbf{R}_e^{\text{cam}}, [t_x, t_y, -1]^\top))$
- 5: Terrain frame ray: $(\mathbf{o}^{\text{ter}}, \mathbf{d}^{\text{ter}}) \leftarrow \text{Transform}^{-1}((\mathbf{p}^{\text{ter}}, \mathbf{R}^{\text{ter}}), (\mathbf{o}, \mathbf{d}))$
- 6: Terrain hit distance: $t^{\text{ter}} \leftarrow \text{RayQuery}(\mathcal{M}^{\text{ter}}, \mathbf{o}^{\text{ter}}, \mathbf{d}^{\text{ter}})$
- 7: Initialize nearest self-occlusion distance: $t^{\text{occ}} \leftarrow +\infty$
- 8: **for** $j = 1$ **to** N^ℓ **do**
- 9: Link frame ray: $(\mathbf{o}_j^\ell, \mathbf{d}_j^\ell) \leftarrow \text{Transform}^{-1}((\mathbf{p}_{j,e}^\ell, \mathbf{R}_{j,e}^\ell), (\mathbf{o}, \mathbf{d}))$
- 10: Link hit distance: $t_j^\ell \leftarrow \text{RayQuery}(\mathcal{M}_j^\ell, \mathbf{o}_j^\ell, \mathbf{d}_j^\ell)$
- 11: $t^{\text{occ}} \leftarrow \min(t^{\text{occ}}, t_j^\ell)$
- 12: **if** $t^{\text{occ}} \leq d_{\min}$ **then break**
- 13: **end for**
- 14: Convert ray distance to camera z -depth: $c \leftarrow (1 + t_x^2 + t_y^2)^{-1/2}$
- 15: $\mathbf{D}[e, h, w] \leftarrow \text{clamp}(c \cdot \min(t^{\text{ter}}, t^{\text{occ}}), d_{\min}, d_{\max})$
- 16: **end for**
- 17: **return** $\mathbf{D}$

D Real-World Deployment Details

We deploy our policy on the AgiBot X2 humanoid robot, which has 29 joints, stands 1.31 m tall, and weighs approximately 39 kg. A forward-facing Intel RealSense D435i depth camera is mounted at the waist and pitched downward by 50° relative to the horizontal plane. The camera provides horizontal and vertical fields of view of 87° and 58° , respectively. Raw 640×360 depth images are filtered, downsampled, and hole-filled using the Intel RealSense SDK, and then cropped to a resolution of 36×36 before being fed into the policy. Depth observations are streamed to the controller at 60 Hz.

For onboard execution, we establish a 1 kHz communication pipeline between the motion control unit (RK3588) and the inference unit (Jetson AGX Orin) via lightweight ROS 2 topic-based messaging. Policy inference runs with ONNX Runtime at 50 Hz and outputs target joint positions for all 21 policy-controlled joints. During deployment, each actuated joint is assigned an individual proportional gain K_p , derivative gain K_d , and action scale, as summarized in Table 14.

Joint Name	K_p	K_d	Action Scale
Hip pitch	120	4	0.25
Hip yaw	100	2	0.25
Hip roll	100	2	0.25
Knee	120	4	0.25
Ankle pitch	40	2.0	0.25
Ankle roll	20	1.0	0.25
Waist yaw	100	3	0.2
Shoulder pitch	30	1	0.2
Shoulder roll	30	1	0.2
Shoulder yaw	30	1	0.2
Elbow	30	1	0.2

Table 14: Joint-wise deployment gains and action scales.

E Cross-Platform Validation

To evaluate the generalizability of SSR across different robotic platforms, we conduct a cross-platform validation on the full-size DEEP Robotics DR02 humanoid. We retain the same training and deployment pipeline used for the primary platform, AgiBot X2, without introducing any platform-specific algorithmic changes. Compared with AgiBot X2, DR02 differs substantially in physical scale and dynamics, standing at roughly 1.8 m tall and weighing around 70 kg. Despite these embodiment differences, SSR transfers effectively to the new platform. As reported in Table 15 and illustrated in Fig. 12, the zero-shot policy achieves high success rates across representative lab-level terrains, while maintaining precise foothold placement and coordinated, natural whole-body motion. These results indicate that SSR is not tightly coupled to a specific hardware configuration, and instead exhibits promising cross-platform transferability across distinct humanoid embodiments.

Terrain	Settings	Success Rate
Stairs Up	15 / 40 cm	100.0% (20/20)
Stairs Down	15 / 40 cm	100.0% (20/20)
Gap	80 cm	100.0% (20/20)
Platform	45 cm	95.0% (19/20)

Table 15: Cross-platform real-world lab-level traversal performance on DEEP Robotics DR02. Success rates are computed over 20 trials per terrain.



Figure 12: Key frames of cross-platform lab-level deployment on the full-size DEEP Robotics DR02 humanoid. The zero-shot transferred policy traverses stairs, an 80 cm gap, and a 45 cm platform, highlighting the cross-platform generalization of SSR.